

Advanced (Carolina Reaper)

Q1. What is the vertex of the parabola $y = x^2 + 4x + 4$?

- (A) $(-2, 0)$
- (B) $(-2, 4)$
- (C) $(0, 4)$
- (D) $(2, 0)$
- (E) $(2, 4)$

Q2. A store is having a sale on bicycles. The price of a bicycle, y , in dollars, is related to the number of bicycles sold, x , by the equation $y = -x^2 + 10x + 50$. What is the maximum price of a bicycle during the sale?

- (A) 75
- (A) 100
- (A) 125
- (A) 150
- (A) 175

Q3. A water tank can hold 120 gallons of water. Due to a leak, the amount of water, y , in gallons, in the tank after x minutes is given by $y = -2x^2 + 20x + 100$. How many gallons of water will be in the tank after 4 minutes?

- (A) 64 gallons
- (B) 80 gallons
- (C) 96 gallons
- (D) 112 gallons
- (E) 120 gallons

Q4. The height of a projectile, y , in meters, is related to the time, x , in seconds, by the equation $y = x^2 - 4x + 5$. What is the axis of symmetry of this parabola?

- (A) $x = 0$
- (B) $x = 1$
- (C) $x = 2$
- (D) $x = 3$
- (E) $x = 4$

Q5. A farmer is planning to build a fence around a rectangular

field. The area of the field, y , in square meters, is related to the length of one side, x , in meters, by the equation $y = x^2 + 2x + 1$. What is the minimum area of the field?

- (A) 0 m^2
- (A) 1 m^2
- (A) 4 m^2
- (A) 9 m^2
- (A) 16 m^2

Q6. A company is analyzing the cost of producing x units of a product. The cost, y , in dollars, is related to the number of units by the equation $y = x^2 - 10x + 50$. What is the minimum cost of production?

- (A) -25
- (B) 0
- (C) 25
- (D) 50
- (E) 100

Q7. The number of cars, y , on a road is related to the time of day, x , in hours, by the equation $y = x^2 - 4x + 5$. What is the y -intercept of this parabola?

- (A) 0
- (B) 1
- (C) 4
- (D) 5
- (E) 9

Q8. The depth of water, y , in meters, in a pool is related to the distance from one end, x , in meters, by the equation $y = x^2 - 2x + 1$. What is the vertex of this parabola?

- (A) $(0, 0)$
- (B) $(0, 1)$
- (C) $(1, 0)$
- (D) $(1, 1)$
- (E) $(2, 0)$

Open Ended Questions (FRQ)

Q9. A skateboarder is performing tricks on a half-pipe. The height of the skateboarder, y , in meters, is related to the horizontal distance from the center of the half-pipe, x , in meters, by the equation $y = x^2 - 4$. Find the maximum height of the skateboarder and the horizontal distance at which it occurs.

Q10. A city planner is designing a new park with a pond. The area of the pond, y , in square meters, is related to the length of one side, x , in meters, by the equation $y = x^2 + 2x + 1$. Find the minimum area of the pond and the length of the side at which it occurs.

Chef's Tips

- Provide clear instructions for each question.
- Emphasize the importance of showing work and explanations for free-response questions.
- Encourage students to use graphs and visual aids to support their answers.